Fiscal Year:	FY 2004	**Task Last Updated:**	FY 01/21/2005
PI Name:	Porter, Marc D Ph.D.		
Project Title:	Multiplexed Colorimetric Solid Phase Extractions: Sensors for Monitoring Spacecraft Water Quality		
Division Name:	Human Research		
Program/Discipline:	ADVANCED HUMAN SUPPORT TECHNOLOGIES		
Program/Discipline--Element/Subdiscipline:	ADVANCED HUMAN SUPPORT TECHNOLOGIES--Advanced environmental monitoring and control		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	mporter@porter1.ameslab.gov	**Fax:**	FY 515-294-3254
PI Organization Type:	UNIVERSITY	**Phone:**	515-294-6433
Organization Name:	Iowa State University		
PI Address 1:	42 Spedding Hall		
PI Address 2:			
PI Web Page:			
City:	Ames	**State:**	IA
Zip Code:	50011	**Congressional District:**	4
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2002 Space Biology 02-OBPR-01
Start Date:	04/30/2003	**End Date:**	09/30/2004
No. of Post Docs:	1	**No. of PhD Degrees:**	1
No. of PhD Candidates:	3	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JPL
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:			
Performance Goal No.:			
Performance Goal Text:			
	This three-year, ground-based project targets the development, ground testing, and KC-135 flight-demonstration of chemical sensor systems and test kit methodologies for application to water quality monitoring in the life support systems used for space exploration. It focuses on the need for enabling analytical technologies that can function as a monitor of spacecraft water quality and as an alarm in the event of contamination. This project seeks to address both the near- and long-term needs of the space program for reliable easy-to-use methods to monitor the levels of key constituents in the drinking water storage distribution systems, including, iodine, silver, lead, cadmium, and nickel. Iodine and silver are the biocides used in the U.S. and Russian systems, respectively. Lead, cadmium, and nickel are contaminants detected in the potable water systems of the International Space Station and must be monitored because of the potential threat to crew health. The unique feature of the proposed system, key components of which were successfully tested in recent KC-135 experiments, draws from the attributes of an		

Task Description:	innovative solid phase extraction (SPE) technique called colorimetric-SPE (C-SPE). C-SPE relies on the immobilization of selective optical reagents, high concentration factors, and the rapid readout of concentrated analytes via a hand held diffuse reflectance spectrophotometer. This work will take this innovation a step further by devising strategies for constructing multiple C-SPE addresses on the same disk. This extension will enhance the ability to rapidly determine several analytes in a set of operational steps, paralleling a multiplexed process. We also plan to employ SPE for the development of a lightweight system that will concentrate organic alcohols (methanol, ethanol, and isopropyl alcohol), propylene glycol, formic acid, and a variety of semi-volatile organic compounds. This “microarchival” concept will dramatically reduce the down mass requirement for archival water quality monitoring. In all instances, efforts will focus on issues of crew health and safety, performance reliability, and minimization of consumables and stowage. To carry out the above tasks, we have assembled a team of scientists and engineers from the Microanalytical Instrumentation Center (MIC) at Iowa State University (ISU) and the Space Water Quality group at the Johnson Space Center (JSC). The ISU team members include Marc Porter (Principal Investigator), James Fritz, Robert Lipert, Dan Gazda, Lisa Ponton, April Hazen-Bosveld, Cherry Shih, and Neil Dias. M. Porter and J. Fritz are on the chemistry faculty at ISU, and R. Lipert is a member of the MIC scientific staff, N. Dias is a postdoctoral fellow, and D. Gazda, L. Ponton, A. Hazen-Bosveld, and C. Shih are graduate students in the ISU chemistry department. The ISU team is charged primarily with the first stage development of the chemical sensor systems and test kits. The JSC team members include John James, John Schultz, Paul Mudgett, Jeffrey Rutz, and Mickie Benoit. J. James is a Chief Toxicologist at JSC. J. Schultz, P. Mudgett, J. Rutz, and M. Benoit are with Wyle Laboratories, a contractor at JSC extensively involved with the Shuttle, Mir, and ISS water quality monitoring programs. J. Schultz coordinates the Wyle efforts. The lead tasks of the JSC team include methods development and validation as well as the design and development of flight compatible hardware for testing in microgravity environments. All stages of hardware design are guided by the extensive experience of the JSC team, and include considerations of system offgassing, reagent toxicity, compatibility with operations in microgravity, and ease of use. The results from our earlier project (NAG9 1191, “Rapid, Specific Chemical Sensors and Test Kit Methodologies for Spacecraft Water Quality Monitoring”) led to the development of a general procedure for the facile, low-level (mid-to-low part per billion) quantification of the biocides Ag+ and I2. The low detection limits largely reflect the high concentration factor (500-1000) obtained by the rapid extraction of an analyte onto a SPE membrane impregnated with an analyte-selective colorimetric reagent. This technology provides a generic platform that has the potential to be readily extended to a host of other analytes important to the space program. The current project takes this innovation a step further by devising strategies for constructing multiple C-SPE addresses on the same disk. The following summarizes the objectives for Year 1. 1. Continue to fine-tune solid phase extraction methods based on disks impregnated with selective colorimetric reagents for biocide (I2 and Ag+) detection by using diffuse reflectance spectroscopy; 2. Develop C-SPE methods for the determination of Ni2+, Pb2+, and Cd2+; 3. Refine protocols of biocide and heavy metal determinations as prototype test kits for KC-135 flight demonstrations and conduct flight demonstrations; 4. Further the development of a microarchival system of spacecraft water contaminants and conduct KC-135 flight demonstrations.; 5. Begin development of a platform for the multiplexed determination of several key spacecraft water quality parameters. The goals of this project reflect the needs of the space program for analytical instruments and methodologies that: 1) meet the monitoring requirements of the International Space Station (ISS) and ongoing Shuttle missions; 2) function as a platform for the development of the analytical methods and ancillary hardware for the projected monitoring requirements for the Mars and other visionary missions; and 3) enable the acquisition of the analytical data necessary for defining further the critical monitoring requirements for crew health and safety in such missions.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The integrated analytical method developed to date allows determination of low concentrations of iodine, iodide and silver ion. A high concentration factor is obtained by extraction of the selected analytes onto an impregnated membrane disk containing embedded polymer particles. No solvent elution or thermal desorption is required because the amount of the colored analyte complex is directly measured on the disk by DRS. The flow-through mode of SPE used here provides a much greater concentration effect and hence better sensitivity than methods in which a color species is measured in solution or on the surface of a probe that is merely dipped into the sample solution. The method is also very rapid. The concentration step requires about 40 s and the reflectance measurement only 2 s. The equipment is small, portable and battery operated so that analyses can be performed on spacecraft in a microgravity environment. The method also provides a generic platform with the potential for extension to the measurement of other analytes, either individually or more excitingly, through the development of a multiplexed C-SPE system for the simultaneous extraction of multiple analytes. Indeed, experiments designed to analyze toxic wastes in ground water (i.e., Ni2+, Pb2+, Cd2+, Cu2+, and the various forms of arsenic) have successfully shown that C-SPE has broad applicability to a wide range of today"s analytical challenges, including pollution monitoring and the rapid, low level detection of explosives and other analytes of important to our Nation"s security.
	1. KC-135 flight demonstrations of the efficacy of C-SPE for use in microgravity. The major goal of the four KC-135 flights this year was to attain agreement between ground and flight data for this methodology. In this set of experiments, in flight water analyses were performed mostly without the added challenge of air bubbles in the test samples. In order to eliminate this difficulty, test solutions were prepared preflight that were devoid of air bubbles. This procedural change allowed us to address whether microgravity is the root cause of the past differences observed between flight and ground-based data. We also sought to test the microgravity compatibility of a method for the determination of nickel. The nickel analysis embodies a new sample manipulation platform and had yet to be tested in microgravity. As such, three of the four flights were aimed at assessing the zero-G performance of the C-SPE determinations of iodine, silver, and nickel.

Another goal was to better characterize the origin of experimental bias between ground and flight data observed in previous KC-135 experiments. Thus, the third flight used bubble-free solutions in syringes and bags would be analyzed along with samples (containing many bubbles) collected from a simulated water system. This comparison would indicate whether the error occurs in filling the Teflon bag from the sample port, from filling a syringe from that bag, or from some other source. The water samples used in these experiments were prepared as follows. Water containing iodine at various levels was obtained by diluting a volumetric iodine standard. These test solutions were analyzed by the LCV method for I2 content. Water containing silver at various trace levels was prepared by diluting a silver standard) with ultrapure water. These test solutions were analyzed by ICP-MS for total and dissolved silver. Dissolved silver was assayed by first filtering the sample or standard with a 0.45 micron nylon syringe filter. Buffered water samples containing nickel were prepared using a nickel standard and sodium borate (pH 9.0, 5% w/w). These test solutions were immediately analyzed by ICP-MS for total and dissolved nickel. Dissolved nickel was assayed by first filtering the sample or standard with a 0.45 micron nylon syringe filter.

All samples were concurrently analyzed by C-SPE both in flight and on the ground. These samples were also analyzed at the same time by the noted baselined laboratory methods. To accurately compare ground and flight C-SPE findings, the following steps were taken: 1) ground analyses were conducted while the aircraft was flying the same experiment; 2) samples, reagents, and all other requisite materials for ground and flight testing came from the same preparative batches; 3) analyses were performed as close as possible to the same temperature; 4) analyses were performed on the ground in the same assembly line fashion as in flight by using a timer to keep each analytical step within the same time intervals as the zero-G intervals obtained in flight; and 5) flights #1, 2, and 4 were done using pre-loaded syringes containing bubble-free samples to eliminate sampling error from bubbles.

Each C-SPE analysis required one parabola to expose the membrane, one to dry the membrane with 60 mL of air, and one to acquire the diffuse reflectance spectrum. The iodine tests were performed using 10.0-mL samples, while the silver and nickel tests both used 1.0-mL samples.

Results - The iodine data showed that the C-SPE analysis is precise, linear, and gravity independent. Plots of both the ground and flight average diffuse reflectance values, expressed as F(R), the Kubelka-Munk function, verses iodine concentration as determined by LCV were linear, with R2 values of 0.999 and 0.997, respectively, with slopes differing by only 7.67%. This agreement is within experimental error. For silver experiments with syringes filled in one-G, both the ground and flight plots of F(R) verses silver concentration determined by ICP-MS were also linear, R2 values of 0.986 and 0.982, respectively, with slopes differing by only 0.63%. For both analyses, this agreement is a marked improvement over previous KC-135 flight experiments, and, more importantly, validates the general utility of C-SPE methodology for operation in microgravity.

For nickel, averaged F(R) for ground and flight measurements were also plotted using the ICP-MS results as the ordinate and the average diffuse reflectance values as the abscissa. Once again, linear regression was used to calculate the best-fit lines shown on the plots. The ground C-SPE data yielded an R2 value of 0.981. The flight C-SPE data, however, exhibited a marked decrease in linearity, having an R2 value of 0.901. The nickel C-SPE data obtained in flight had a slope that was half that of the ground C-SPE data, and relative standard deviations that were much higher than the ground data. By inspection, it was clear that the nickel flight membranes failed to develop a color intensity close to that observed on the ground. We also found that much more of the water sample remained upstream of the membrane upon disassembly of the cartridge. These discrepancies indicate an inability to effectively meter the 1-mL sample volume through the cartridge used in the nickel determinations, which had a much larger dead volume than that used in the silver and iodine analyses. We note that this shortcoming was revealed as a result of the microgravity simulation, demonstrating the importance of continued KC-135 testing. We are presently examining approaches to markedly reduce this dead volume.

In Flight #3, the C-SPE results obtained from the two preloaded syringes at each silver concentration produced a linear plot (R2 =0.992) with a slope that agrees with the results from the fourth flight. Five silver analyses each were done from a sample bag that had been degassed. Five additional samples were analyzed after collection from an identical bag that had been sampled from a water system simulating the space station water system, which allowed gas bubbles to enter the bag during sampling. Comparison of the precision and accuracy of these two sets of data was to indicate which step(s) of the procedure introduces the error caused by air bubbles. The bubble-free bag averaged an F(R) of 0.0641. Substituting this value into the linear equation derived from the silver standards analyzed during this flight gives 0.42 mg/L, or 84.7% recovery on this 0.50 mg/L standard. The bag sampled with bubbles present gave a value of 0.35 mg/L, also on a 0.50 mg/L standard. The relative standard deviation increased from 16% for pre-filled syringes to 29% with the addition of bubbles. This result clearly mandates the need for bubble removal, which is a key focus in the next phase of our research effort.

Another important change in our planning for the KC-135 flights should be noted. For this set of flights, we invoked a more rigorous preflight training program by extensively rehearsing each analysis on the ground the full week before the actual flights. With this change, all flyers repeatedly practiced all facets of each experiment. During this process, operational issues and details were carefully examined, with several improvements in each procedure identified and implemented. This procedural change clearly enhanced the effectiveness of the flight-testing, and will be adopted for all future simulations.

We add that the larger deviation for the silver determinations with respect to that for the iodine determinations reflects another important procedural change which we will put in place. All iodine preparations on-ground were conducted by a single flyer, whereas those for silver involved several members of our flight team. We therefore attribute the larger deviation for silver to subtle differences in how individuals perform the same task. We are presently working on procedures and training documents to address this issue. Work is also underway to devise a bubble removal procedure, which is viewed as the principal obstacle to onboard deployment.

Conclusion- These KC-135 experiments have validated the effective operation of C-SPE as a method for the quantification of biocides in water samples in reduced gravity. The iodine C-SPE method was the most precise, showing close agreement between ground and flight results. The silver C-SPE method proved to be feasible with a drinking water sampling system almost identical to that currently used on the space station now. C-SPE was shown not to be effective in the analysis of nickel in its current configuration for in flight deployment. However, the dead volume shortcoming should be readily correctable through a redesign of fluid handling hardware. Overall, the results of these experiments show that, as a water quality monitoring system, C-SPE has the potential to provide a much-needed analytical capability for operation on the ISS.

Task Progress:

2. Multiplexed Colorimetric Solid Phase Extraction (MC-SPE)

The creation of MC-SPE, which represents a new addition to parallel analysis systems, was driven in large part by the limited astronaut time available on ISS to conduct water testing. As such, the development of a multiplexed water analysis system has the potential to increase the collection of water quality data with a minimal investment of on-orbit crew time. In extending C-SPE to a multiplexed format (MC-SPE), a filter holder that incorporates discrete analysis channels and a jig that facilitates the concurrent operation of multiple syringes have been designed, enabling the simultaneous determination of three different measures of water quality. Separate, single parameter membranes, placed in a read-out cartridge create unique, parameter-specific addresses at the exit of each channel. Following sample exposure, the diffuse reflectance spectrum of each address is collected serially and the Kubelka-Munk function, F(R), is used to quantify each water quality parameter via calibration curves. Performance evaluations of the MC-SPE platform were conducted using sample pH, silver(I), and nickel(II). Determinations of silver(I) (0.05-0.5 ppm) and nickel(II) (1.8-5.0 ppm) follow established C-SPE methods on reversed phase extraction membranes using 5-(p-dimethylaminobenzylidene) rhodanine and dimethylglyoxime, respectively, as colorimetric reagents. Sample pH (2.5-5.0) is measured using an anion exchange membrane impregnated with fluorescein. These determinations require ~120 s to complete using a total sample volume of 3.0 mL.

A syringe jig was constructed that facilitates the concurrent operation of multiple syringes. The jig positions the syringes to allow all three plungers to be depressed simultaneously, introducing sample to the multiplexed filter holder. The multiplexed filter holder is composed of four sections, a syringe interface, a flow chamber, a read-out cartridge, and a base. Fabrication of each section is described below.

Syringe interface. The upper portion of the filter holder directs the sample flow from each syringe to one of the three analysis channels in the flow chamber. This piece was fabricated from 35-mm diameter white delrin. The bottom of the interface was machined to mate directly to the flow chamber, leaving no dead volume. Samples flow through the syringe interface into the flow chamber via three 1.6 mm-diameter ports lined with PEEK tubing. The PEEK port lining protrudes from the top of the interface and is connected to nylon Luer slip fittings with Silicone tubing. The Luer fittings were pressed into an acrylic brace to simplify connection to the sample syringes.

Flow Chamber. The center portion of the filter holder defines each analysis channel. The flow chamber was fabricated from 35-mm diameter white delrin cut to a length of 33.3 mm. Three equally spaced 7.0 mm-diameter holes were drilled along the axial direction of the cylinder to create the three analysis channels. The top of each channel was bored out to accept a 9.0-mm diameter support screen. These screens rest 12.0 mm above the channel outlets and facilitate the placement of the wool and paper used for the in-line addition of immobilized reagents (nickel(II) method) in the channels. The bottom of the flow chamber was machined to fit over the base of the filter holder. Base. The bottom portion of a 25-mm diameter Swinnex polypropylene filter holder was used as the filter holder base.

Read-out cartridge. Two 25-mm diameter circles were cut from 0.05-mm thick nitrile rubber. These circles were stacked and sealed together at one point on their circumference with double-sided tape. Three equally spaced 7-mm diameter holes were cut through the rubber circles to align with the analysis channels in the flow chamber and define the area of the read-out addresses. Loading individual, parameter-specific membrane disks between the two layers of nitrile rubber creates a single read-out cartridge with three separate 7.0-mm addresses.

Clamp. The entire multiplexed filter holder is held together with a brass screw clamp. The clamp consists of two brass rings machined to fit snugly around the syringe interface and base of the filter holder. Following assembly of the filter holder (described below), three screws are inserted through the ring at the syringe interface and threaded into the ring at the holder base. Tightening these screws secures the filter holder sections in place.

Three 10-mm diameter C-SPE membrane disks, one specific for each sample parameter, were loaded in the read-out cartridge. The read-out cartridge was placed on the filter holder base and the flow chamber was positioned such that the outlets of the analysis channels were aligned with the addresses on the read-out cartridge. Next, support screens were inserted in each flow channel. A DMG chad and a circle of buffer wool were added, in that order, to the support screen in the analysis channel above the nickel(II) membrane. The sample interface was positioned on top of the flow chamber ensuring that the PEEK-lined ports were oriented at the inlet of the analysis channels. Finally, the entire assembly was clamped together using the brass screw clamp.

A mask was used to match the aperture size of the spectrophotometer to the diameter of the addresses on the read-out cartridge. The mask was made by cutting a 7-mm diameter hole in a 100-µm thick polyester sheet and then painting the sheet flat black with spray paint. The mask was affixed to a BYK-Gardner sample locator using double sided tape.

Sampling procedure. Three 1.0-mL aliquots of a water sample were collected in 1.0-mL plastic syringes. These syringes were fitted into the syringe jig and connected to the syringe interface of the filter holder. The plungers of all three syringes were depressed simultaneously to meter 1.0 mL of sample solution through each analysis channel in the flow chamber. Following passage of these solutions, the syringes were removed and the membranes were dried by passing air through the filter holder with a 60-mL syringe and an adapter mated to the analysis channels. At this point, the filter holder was disassembled and the read-out cartridge was removed. The cartridge was placed under the sample locator so that one read-out address was visible through the aperture mask. The spectrum of that address was acquired and the cartridge was repositioned to sample the other two addresses in a serial fashion. This process entire process, beginning with syringe filling, requires ~120 s. Results-pH method development. There are three main reasons FL was chosen to demonstrate the ability to measure pH optically in MC-SPE. First, the biodegradability and low toxicity of FL make it an ideal reagent for use in spacecraft environments. Second, experiments have shown that the absorbance of aqueous FL at 490 nm changes with pH in the range 1.25 to 8.70. And third, most common metal ions do not interfere with the dye response.

The response of the FL membranes was assessed by running conventional C-SPE trials using 13-mm diameter membrane disks and 0.1 M buffer solutions with pH values between 2.17 and 6.23 as samples. The results of these experiments were compared to the published absorbance studies to determine if immobilization affected the reactivity of the dye. Because the diffuse reflectance spectrophotometer records relative reflectance in 20-nm intervals, the relative reflectance of the samples at 490 nm could not be recorded. Instead, the value of F(R) calculated from the relative reflectance at 480 nm was used to compare C-SPE results to absorbance experiments. When 1.0-mL buffer samples were tested, the trend in the plots of F(R) at 480 nm and the absorbance at 490 nm versus pH were very similar. However, because of the large absorbtivity of the FL dianion, the relative reflectance values above pH 4.5 are in the high

error regime of spectrophotometric measurements. In order to reduce possible errors in pH measurements made by C-SPE, a detection wavelength where the pH membranes have a higher relative reflectance, and therefore a lower F(R) value, was used. Based on a visual inspection of the data, the F(R) value at 560 nm was selected. The F(R) values at this wavelength follow the same trend as the data at 480 nm, enabling determinations of sample pH between 2.17 and 6.23 with reduced error.

Multi-analyte experiments. To validate the performance of our MC-SPE system, calibration plots were constructed for silver(I), nickel(II), and pH using the three-channel filter holder. The collection of the silver(I) and nickel(II) data was accomplished by loading all three addresses in the read-out cartridge with C-SPE membrane disks specific for one analyte and running standard solutions containing only that analyte through the multiplexed holder. In the case of nickel(II), a DMG chad and a circle of buffer wool were loaded in each analysis channel before sampling. The F(R) values collected in these experiments were then averaged and plotted against the analyte concentration to obtain calibration plots. Data for the pH calibration were gathered using the silver(I) and nickel(II) standard solutions and a read-out cartridge loaded with three pH sensitive membrane disks. The F(R) values obtained from these trials were plotted against the pH values of these standards, as measured with a pH meter.

Silver(I) calibration. The same wavelength of detection used in the conventional C-SPE experiments, 580 nm, was employed for all silver(I) measurements with MC-SPE. These determinations exhibited an upper limit of the linear response range near 0.5 ppm. Above this concentration, the F(R) values at 580 nm begin to level off. The lower limit of detection, calculated as the blank signal plus three times the standard deviation of the blank was 0.05 ppm.

The observed leveling of the response is attributed to saturation of the DMABR. Conventional C-SPE measurements, which use a 13-mm diameter membrane disk to extract analytes also exhibit saturation, but at a higher level of silver(I) (i.e., ~1.0 ppm). The addresses on the read-out cartridge are only 7 mm in diameter, representing an active area that is ~70% smaller than that used in our conventional (single analyte) C-SPE system. Since both C-SPE platforms use 1.0-mL sample volumes, the multiplexed membrane disk saturates at a lower concentration than the conventional C-SPE disk. In the linear response region, between 0.05 ppm and 0.5 ppm, all measurements showed an uncertainty of less than 15%.

Nickel(II) calibration. The same wavelength of detection used in the conventional C-SPE experiments, 540 nm, was employed for all nickel(II) measurements with the multiplexed filter holder. Unlike the silver(I) experiments, the nickel(II) studies performed with the multiplexed holder demonstrated a smaller linear response range than the single analyte method. This decrease cannot be associated with membrane disk saturation, as the upper concentration limit of the analysis has been shown to be limited by the release of DMG from the reagent chad. Rather, the decreased range is a consequence of an increase in the concentration limit of detection from 0.5 ppm to 1.8 ppm.

The loss of membrane disk signal at low concentrations is ascribed to two factors: the proximity of the buffer wool to the reagent chad and the decrease in dead volume between the reagent chad and the collection disk. Both factors reduce the reaction time for the sample before passing through the membrane disk. In the conventional nickel(II) platform, the sample syringe is filled through a buffer wool cartridge. This step allows the sample to equilibrate at pH 9.0 in the syringe barrel while the buffer cartridge is detached from the syringe and the reagent chad/collection membrane assembly is attached to the syringe. In the multiplexed holder, the buffer wool sits directly on top of the reagent chad. As a consequence, solutions passing through the wool immediately encounter the reagent chad, which reduces the time available for equilibration of the buffer and sample before the sample encounters the reagent chad. Our earlier report on this method showed that if the sample does not reach the proper pH, the rate of reaction is severely diminished and the precipitate may not exhaustively form before the solution passes through the read-out address. This loss of precipitate will lower the F(R) at all concentrations, and have a negative effect on the limit of detection.

Decreasing the dead volume between the reagent chad and the collection address should have a similar effect on the measurement. In the conventional method, once the solution passes through the reagent chad, it fills the dead volume between the chad and the collection membrane disk. This dead volume is slightly less than the total sample volume (1.0 mL), so a small portion of the sample is passed through the membrane disk immediately. The remainder of the sample sits stationary above the collection disk until it is displaced by air during membrane drying. The time lapse associated with the removal of the sample syringe and connection of the drying syringe allows more time for the analyte and reagent to complex in solution. In the multiplexed holder, this dead volume is roughly half (~0.4 mL) that in the two cartridge assembly used for conventional experiments (~0.8 mL), but the sample volume is the same. As a result, more of the sample passes through the collection address without the benefit of extra residence time. At lower concentrations, samples may not form enough precipitate to be distinguished from the background. Despite the increased limit of detection, measurements made in the linear response region for nickel(II) in the multiplexed holder, 1.8 to 5.0 ppm, show an uncertainty similar to that of the silver(I) calibration, less than 15%.

pH calibration. In initial experiments with the silver(I) and nickel(II) standards, the response from the membrane disks was somewhat erratic, failing to follow the trend observed with the buffer solutions. This behavior was attributed to the varying ionic strengths of the standards, which has a known effect on the response of pH indicators, including immobilized forms of FL. The pH platform was tested again using silver(I) and nickel(II) standard solutions prepared in 0.05 M sodium nitrate to adjust the ionic strength of the standards. The response from these new solutions, which had measured pH values between 2.5 and 5.0, was more reproducible and showed an uncertainty less than 17%. Additionally, the demonstrated relation between F(R) at 560 nm and pH agreed closely with that observed using buffer samples. We note that the FL-impregnated membranes exhibited a response between pH 2.17 and 6.23, but our calibration run was restricted to the pH range encompassed by the silver(I) and nickel(II) standards.

The response of the pH membrane disks deviates from the sigmoidal behavior typically exhibited by pH indicators in absorbance studies. This same deviation is found in the solution absorbance studies with FL and has been attributed to the fact that all four forms of FL contribute to the solution absorbance at 490 nm. Because F(R) at 560 nm follows the same trend with pH as absorbance studies, multiple forms of FL are clearly contributing to the C-SPE response.

Multi-analyte trials. The sample solutions used in the performance evaluation of MC-SPE were prepared to contain levels of silver(I) and nickel(II) that spanned the linear response range of the two metals as defined by the calibration studies. Each sample was run in triplicate (N=3) and the average F(R) value at the wavelength of detection was used for quantification of all sample parameters.

Overall, there is strong agreement between the multiplexed determinations and the actual values of the sample parameters. All but one measurement showed a percent difference less than 14%, and all but three demonstrated a

	difference below 10%. Several measurements were discarded because of visible misalignment of the membrane disks with the addresses in the read-out cartridge. The cause of the misalignment is thought to be a shifting of the membrane disks between the layers of the read-out cartridge during assembly of the multiplexed filter holder. This created a small gap that allowed a portion of the sample to leak around the membrane in the read-out address. This issue will be addressed in future designs of the multiplexed platform by simplifying holder assembly and minimizing the need for manual alignment of platform components. Importantly, these results validate the performance of our MC-SPE platform for the concurrent analysis of several water quality parameters. Conclusions. The ability of a MC-SPE to simultaneously determine three of NASA's Priority 1 water quality parameters has been demonstrated. Methods encompassing two variations of C-SPE and requiring chemically different membranes were successfully combined to measure silver(I) levels ranging from 0.05 to 0.5 ppm, nickel(II) levels between 1.8 to 5.0 ppm, and sample pH between 2.5 and 5.0. All three parameters were determined in ~120 s using a total sample volume of 3.0 mL. Future work with MC-SPE will focus on two areas, improving the present hardware and additional applications of the multiplexed platform. Filter holder designs that provide results covering the concentration ranges required by NASA, simplify assembly, minimize alignment, and reduce the possibility of leakage at the read-out cartridge are planned. Also, a method whereby a single sample syringe can be used for sample introduction is being developed. Extensions of MC-SPE to the speciation of aqueous iodine solutions and possible use in the extraction of trace organics from water samples are being explored. Research to enhance the range of the pH method through the co-impregnation of multiple indicators on a membrane as well as application of the multiplexed platform to meet environmental monitoring needs on Earth, such as the determination of lead and arsenic is also underway.
Bibliography Type:	Description: (Last Updated: 03/31/2017)
Articles in Peer-reviewed Journals	Gazda DB, Lipert RJ, Fritz JS, Porter MD. "Investigation of the iodine-poly(vinylpyrrolidone) interaction employed in the determination of biocidal iodine by colorimetric solid-phase extraction." Analytica Chimica Acta, 2004 May;510(2):241-7. http://dx.doi.org/10.1016/j.aca.2004.01.010 , May-2004
Articles in Peer-reviewed Journals	Gazda DB, Fritz JS, Porter MD. "Determination of nickel(II) as the nickel dimethylglyoxime complex using colorimetric solid phase extraction." Analytica Chimica Acta, 2004 Apr;508(1):53-9. http://dx.doi.org/10.1016/j.aca.2003.11.044 , Apr-2004
Articles in Peer-reviewed Journals	Gazda DB, Fritz JS, Porter MD. "Multiplexed colorimetric solid-phase extraction: determination of silver(I), nickel(II), and sample pH." Anal Chem. 2004 Aug 15;76(16):4881-7. http://dx.doi.org/10.1021/ac049449r , Aug-2004
Papers from Meeting Proceedings	Dias NC, Gazda DB, Fritz JS, Porter MD, Rutz J, Mudgett P, Schultz J. "Application of Colorimetric Solid Phase Extraction (C-SPE) to Monitoring Nickel(II) and Lead(II) in Spacecraft Water Supplies." 2004 International Conference on Environmental Systems, Colorado Springs, CO, July 2004. 2004 International Conference on Environmental Systems, Colorado Springs, CO, July 2004. ICES paper 2004-01-2539. http://dx.doi.org/10.4271/2004-01-2539 , Jul-2004
Papers from Meeting Proceedings	Ponton LM, Gazda DB, Lipert RJ, Fritz JS, Porter MD, Rutz J, Mudgett P, Dungan D, Schultz J. "Archiving Trace Organic Contaminants in Spacecraft Water." 33rd International Conference on Environmental Systems, Vancouver, B.C., Canada, July 2003. 33rd International Conference on Environmental Systems, Vancouver, B.C., Canada, July 2003. SAE Technical Paper 2003-01-2408. http://dx.doi.org/10.4271/2003-01-2408 , Jul-2003
Papers from Meeting Proceedings	Gazda DB, Lipert RJ, Fritz JS, Porter MD, Rutz J, Mudgett P, Schultz J. "Rapid Determination of Biocide Concentrations Using Colorimetric Solid Phase Extraction (C-SPE): Results from Microgravity Testing." 33rd International Conference on Environmental Systems, Vancouver, B.C., Canada, July 2003. 33rd International Conference on Environmental Systems, Vancouver, B.C., Canada, July 2003. SAE Technical Paper 2003-01-2406. http://dx.doi.org/10.4271/2003-01-2406 , Jul-2003